Alexey A. Tatarinov

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Objective

4th year undergraduate electrical engineering student seeking an internship in robotics and manufacturing starting May 2027. 3+ years of expertise in labs and makerspaces in building and testing analog and embedded electronics systems. Successes include projects ranging from hardware and firmware for nanosatellites to layout and validation of precision amplifier and filter PCBs, and firmware for control systems.

Education

Georgia Institute of Technology | Atlanta, GA *August 2023 – Present*
Bachelor of Science in Electrical Engineering, GPA 3.91 Expected Graduation May 2027
Concentrations in Circuit Technology and Robotics & Autonomous Systems

Skills

Programming: Java, Python, ISO/ANSI C, C++, Rust, JavaScript, TypeScript, Julia, MATLAB
Platforms: UNIX/Linux (Ubuntu, Slackware, Debian), Windows, RTOS (Embassy, Zephyr), HDL (System Verilog, VHDL)
Hardware: Raspberry Pi, ARM MCUs, FPGA RTL, ESP32, oscilloscope, logic analyzer, power supplies
Software: Altera Quartus II, NI LabVIEW, CAD (OnShape), GitHub, SPICE (LTSpice, NGSpice), EDA (Altium, KiCad)
Communication: Design proposals, technical reports, instruction manuals, presentations (large and small audiences)
Languages: English (fluent), Spanish (beginner)
Professional Organizations: Eta Kappa Nu, IEEE

Experience

Systems Control at Texas Instruments | Dallas, TX *May 2026 – July 2026*
Control Systems Engineering Intern (40 hours/week)

- Built KPI data analysis framework using data-driven techniques (principle component analysis, information criteria), finding significant differences between production planning engine runs with the same inputs that could cause chaotic system behavior.
- Implemented dispatch level supply demand and inventory simulator supporting timed capacity constraints across 1000s of materials using linear programming and statistical analysis, that was used to evaluate strategies for using excess capacity.
- Wrote a logpoint level manufacturing simulator in Rust, allowing simulating 1000s of materials with near instant feedback, using a greedy scheduling algorithm that allows dynamic adjustments.

Electrical and Computer Engineering at Georgia Institute of Technology | Atlanta, GA *January 2025 - May 2026*
Undergraduate Research Assistant (10 hours/week)

- Guided the schematic design, assembly, and debugging of 3 low noise precision amplifier PCBs with active feedback using signal integrity principals, which were critical to demonstrating feasibility of proposed acousto-electronics research.
- Used ECG monitors as source of voltage data to calibrate measurement tools with another source, ensuring robust operation with different data sources.
- Collected vibrometer data used to build models of guitar acoustic behavior to determine locations for pickup placement.
- Designed and implemented composable second order filter board with high Q notch and Butterworth low/high pass filters, doubling SNR in measured data.
- Tested circuits using oscilloscopes with frequency response analysis, ensuring design specifications were met.

Physics Department at Georgia Institute of Technology | Atlanta, GA *August 2025 – December 2025*
Undergraduate Teaching Assistant (10 hours/week)

- Hosted weekly office hours for 20 student course, including a graduate student section, helping students understand the material better and catch minor errors.
- Graded over 100 homework submissions, giving thorough feedback on each part, receiving positive feedback from several students noting how it improved their learning.

Tutoring and Academic Support at Georgia Institute of Technology | Atlanta, GA *January 2025 - April 2025*
Peer Lead Undergraduate Study Leader (10 hours/week)

- Designed twice weekly lecture review sessions for a calculus-based electromagnetics course including interactive elements and group work that drew an average of 15-25 attendees.
- Prepared and led 2-hour exam review sessions drawing over 200 students for each exam, providing them an additional chance for group review before last minute studying.
- Met weekly with other PLUS leaders to discuss effective teaching techniques and how to create welcoming environment.

ASCEND Climbing | Pittsburgh, PA *June 2024 – August 2024*
Youth Assistant Coach (~10 hours/week)
- Taught weekly groups of 15 kids ages 12-16 about climbing principles and techniques including belaying, rappelling and lead climbing through interactive demonstrations, making kids more enthusiastic about climbing.
- Facilitated conversations about teamwork/safety ideas including accountability, optimism, and persistence keeping the children motivated and safe.

Electrical and Computer Engineering at Carnegie Mellon University | Pittsburgh, PA *October 2022 – August 2023*
Intern (10+ hours/week)
- Implemented graph algorithms operating on CSR form matrices in C to benchmark power efficiency against parallel dataflow architecture, showing advantages for energy constrained environments.
- Created mesh communication system architecture to allow robust communication between router hardware on thousands of nanosatellites, enabling constellation computing without modifying the existing protocol.
- Wrote firmware for stacked nanosatellite design involving STM and nRF microcontrollers, including custom bootloader allowing wireless uploads of arbitrary programs crucial to actual deployment.

SigmaCamp | Silver Lake, CT *October 2022 – Present*
Counselor, Teaching Assistant, Problem of the Month Lead
- Assisted with week-long seminars that taught students ages 11-17 subjects including dynamic equations and control systems, semiconductor manufacturing, robotics, and machine learning/data science, resulting in campers academic enrichment on subjects not usually covered before college.
- Devised engineering and linguistics problems for annual online competition, bringing in over 100 competitors each year.

Sarah Heinz House | Pittsburgh, PA *October 2021 – June 2022*
Robotics Teaching Assistant
- Taught a class of 15 students about STEM concepts focused on robotics using basic physics principles, promoting interest in robotics and higher-level STEM programs at Sarah Heinz House.
- Designed 2 STEM lessons involving interactive demonstrations and challenges, teaching how to program sensors for improved path finding, and how to apply kinematics to a zipline.

Projects

NFFT Guitar Analysis *Summer 2025*
Microelectronic Acoustics Group, Georgia Tech Electrical and Computer Engineering
Individual project to better understand previously collected vibrational data from an acoustic guitar.
- Coded high performance 3D data visualization with Vulkan Scene Graph allowing vibration data to be analyzed and compared with various statistical models.
- Applied JSON serialization to simplify interop between various components, allowing more understandable implementations using tools better suited for each domain.
- Made Python code to match data driven models and identify system dynamics, establishing nonlinear behavior through identifying trends and flaws.

Advanced LED Controller
Digital Design Lab *Spring 2025*
Group project to apply the engineering design process and digital design principles.
- Led group brainstorming and project management, making sure every group member contributed to the project.
- Followed the engineering design process to identify, prototype, draft, and refine a working FPGA product.
- Implemented the majority of the project code using VHDL to create a peripheral with multiple registers to allow for customizable control including PWM based dimming and composable animations/patterns, resulting in a useful device.

- Drafted thorough documentation for end-users including basic validation steps and troubleshooting for common issues, improving the development experience.

Relevant Coursework

Control System Design: State-space notation and analysis of linear systems; Modal control and stabilization; Digital implementation of state space control methods; Interrupt-based embedded programming and application design.
Senior Analog Laboratory: Transistor design of amplifiers, sources, and op amps; Measurement of semiconductor characteristics and non-idealities; Analysis of transistor circuits using Hybrid-Pi and T models.
Signal and Systems: System analysis in the Fourier, Laplace and Z domains; Bode and Nyquist plots; Linearization of nonlinear systems; Radar design; MATLAB; PID control.
Measurements, Circuits and Microelectronics Laboratory: Circuit analysis techniques: impedance, transconductance, transfer functions; Analysis using oscilloscope, function generator, multimeter.
Microelectronic Circuits: Semiconductor materials and properties; small-signal modeling and analysis; MOSFET and BJT equations and circuits.
Digital Design Laboratory: System validation using logic analyzer; Digital logic and FPGA simulation/synthesis.

Leadership

The Hive Makerspace | Atlanta, GA
Director of Strategy (10+ hours/week) *May 2026 – Present*
- Organized and ran weekly executive board meetings handling outreach, policy and technical strategy, improving the interview and onboarding processes, resulting in higher application counts and event turnouts.
- Launched collaborations with other student organizations on campus, including a hackathon reaching over 1000 students.

Peer Instructor (5+ hours/week) *December 2024 – May 2026*
- Staffed space serving over 1000 students every semester helping with soldering, power supplies, 3D prints, PCBs, and shop tools, supporting 100s of student projects and several research groups.
- Ran workshops reaching over 20 students on subjects including circuit design/assembly and sublimation and laser cutting, promoting tools in the space and letting students develop creative skills.

Georgia Tech Sailing Club| Atlanta, GA *April 2024 – Present*
Treasurer (5+ hours/week)
- Managed over $40,000 in annual club spending, allocating funds across race team travel, organization fees and maintenance of a fleet of boats worth over $30,000.
- Drove changes to club bylaws to simplify travel expenses, reducing the burden on the executive board and opening new possibilities for events and regattas.
- Built a new expense tracking system that cut audit times from several hours to under 30 minutes and allowed greater introspection into spending.